STAFF REVIEW · H446 053 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. In the left subtree.

2. Node representing a point/entity.

3. A queue is first in, first out (FIFO); a stack is last in, first out (LIFO).

4. Row C is [1,0,0,1], because C connects to A and D. For each undirected edge, both corresponding matrix cells are 1.

I Do — source worked steps

1. Explain the trace — With B before C, BFS visit order A,B,C,D. Each vertex is marked on enqueue.

We Do — guided CHECK

1. Explain the trace
BFS visit order X,Y,Z,W. FIFO queue processes all distance-one vertices before W.

You Do — source mark schemes

1. A:[B,C] [2], B:[A,D] [2], C:[A],D:[B] [2].

2. Start [A] [1], after A [B,C] [2], after B [C,D] [2], visited contains A,B,C,D after discovered [1].

3. DFS LIFO deep path [2], BFS FIFO frontier [2], visited prevents repeated processing/infinite revisits [2].

Exit Ticket — source answer

1. False. Both A and B list one another.